



Aishu R 251001007-2025-IT ▾

A2**Started on** Tuesday, 4 August 2026, 9:00 AM**State** Finished**Completed on** Tuesday, 4 August 2026, 9:01 AM**Time taken** 33 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  void function(int n)
3  {
4      int c= 0;
5      c=c+1;
6      for(int i=n/2; i<n; i++){
7          c=c+2;
8          for(int j=1; j<n; j = 2 * j){
9              c=c+2;
10             for(int k=1; k<n; k = k * 2){
11                 c=c+2;
12             } //c=c+1;
13         } //c=c+1;
14     }
15     c++;
16     printf("%d",c);
17 }
18 int main(){
19     int n;
20     scanf("%d",&n);
21     function(n);
22     return 0;
23 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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